

Project Goal

To design a solution that lowers the incidence rate of violence against healthcare workers in the workplace, improves the security response time to violent incidents, and requires minimal training to implement and use.

Background

Workplace Violence in Hospitals

- Healthcare workers are five times more likely to experience violence in the workplace than workers in any other industry
- The rate of injury from violent attacks on healthcare workers has increased 63% from 2011 to 2018 [1]
- Most violence experienced by healthcare workers is verbal

Hospital Access and Security

- Historically, obtaining access to a hospital has been relatively easy, with many access points that were largely unsecured and open
 - Now, entrances to hospitals have become more secure, with many security teams utilizing metal detectors and weapons screening
- There has been a shift to using more technology in the management of safety and security in hospitals in response to the lack of available resources, both financially and with staffing shortages [2]

Computer Vision/Tracking Currently Being Deployed

- Computer vision is already being used for security purposes
 - Track location of workers
 - Facial recognition
 - Emotional analysis

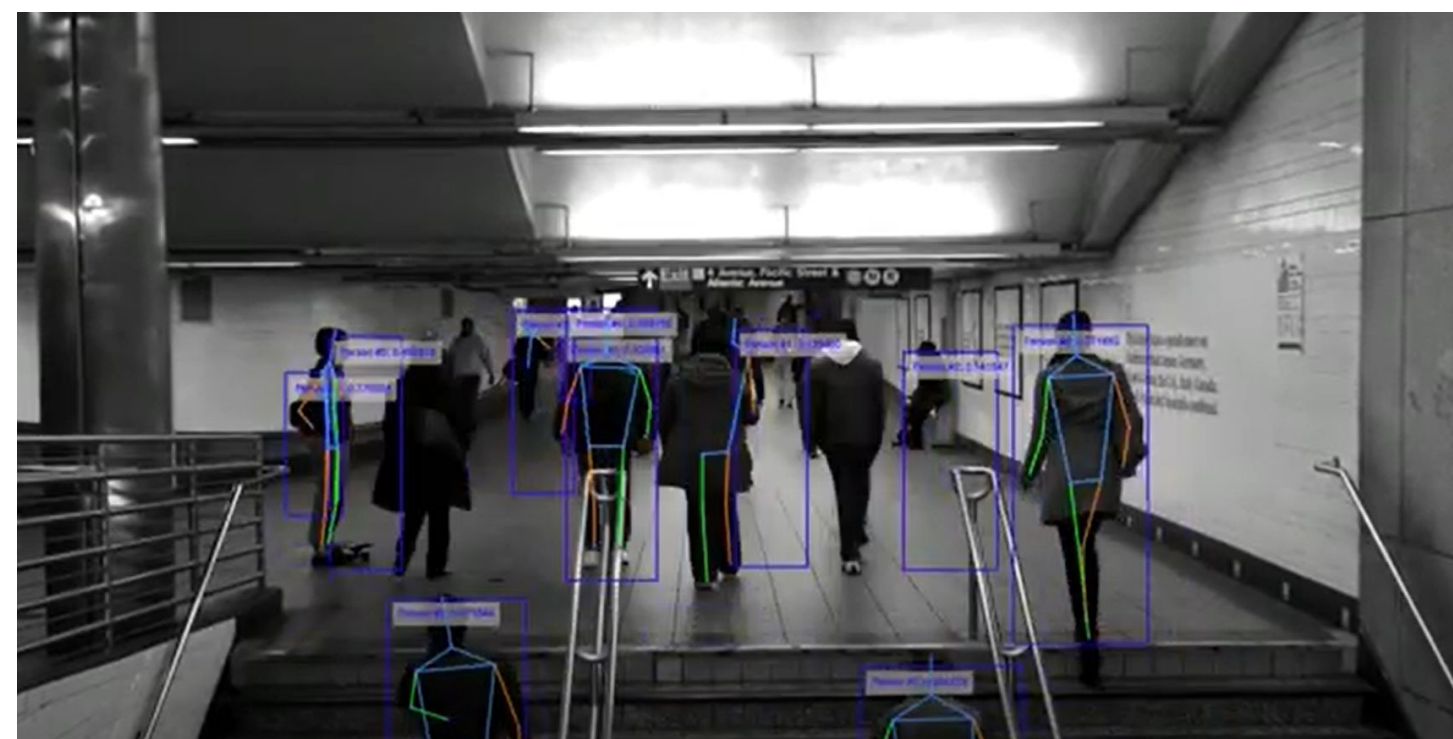


Figure 1. Example from Ikomia of AI in security.

Methods and Design

- Use computer vision and audio to determine the chance of an incident
- Initially tried pose detection using Open Pose
 - Abandoned due to algorithm speed and tracking difficulties
- Switched to object detection and tracking using YOLO v8
- Parameters collected from video and audio inputs for Safety Factor calculation:

- Number of People
- Speed of People (Activity)
- Noise Level
- User set parameters:
 - People Threshold
 - Speed Limit
 - Activity Sensitivity
 - Sound Sensitivity
 - Sound Limit

- All parameters were weighted and used to calculate a Safety Factor that would be used to determine the potential for a violent incident

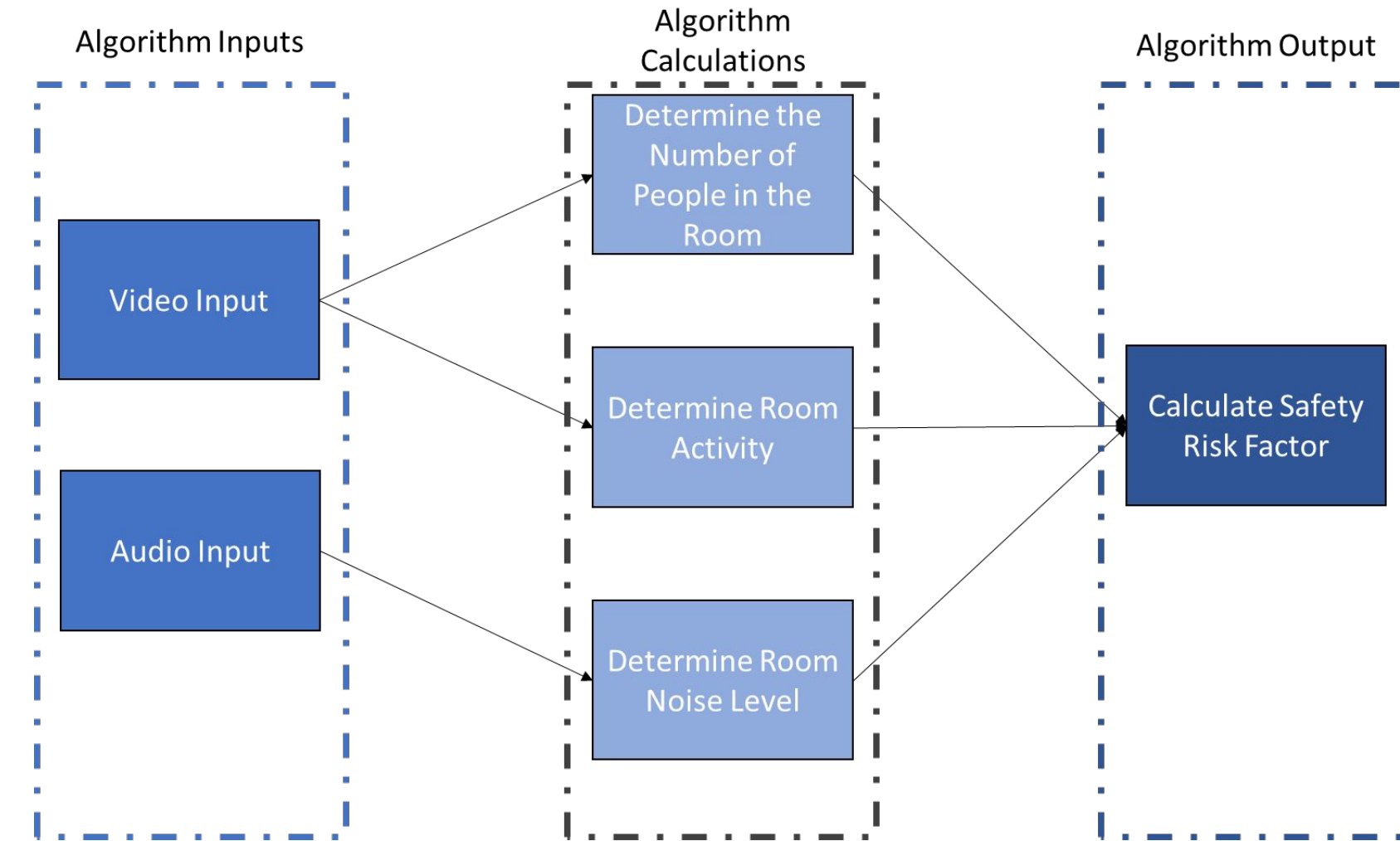
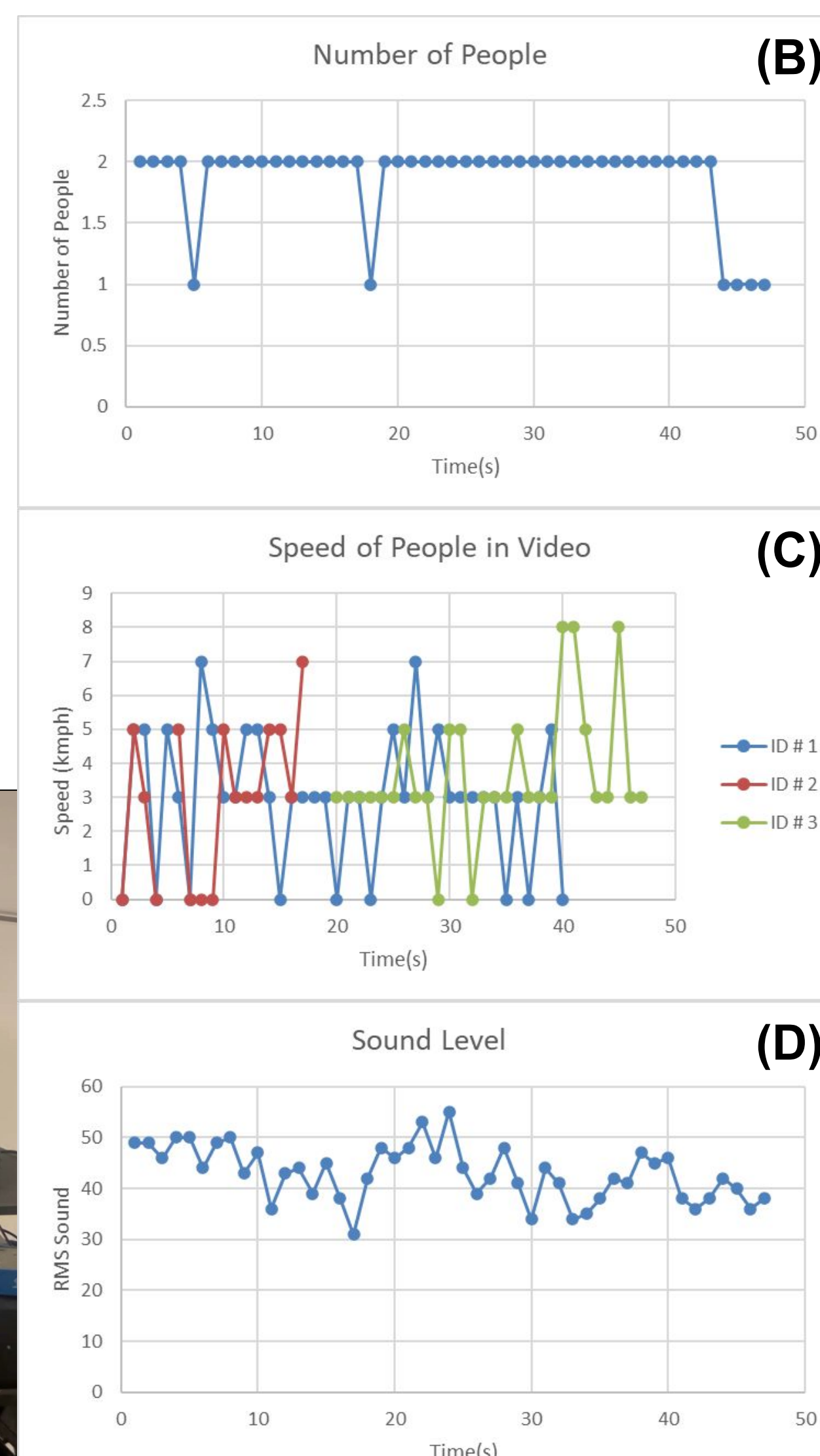
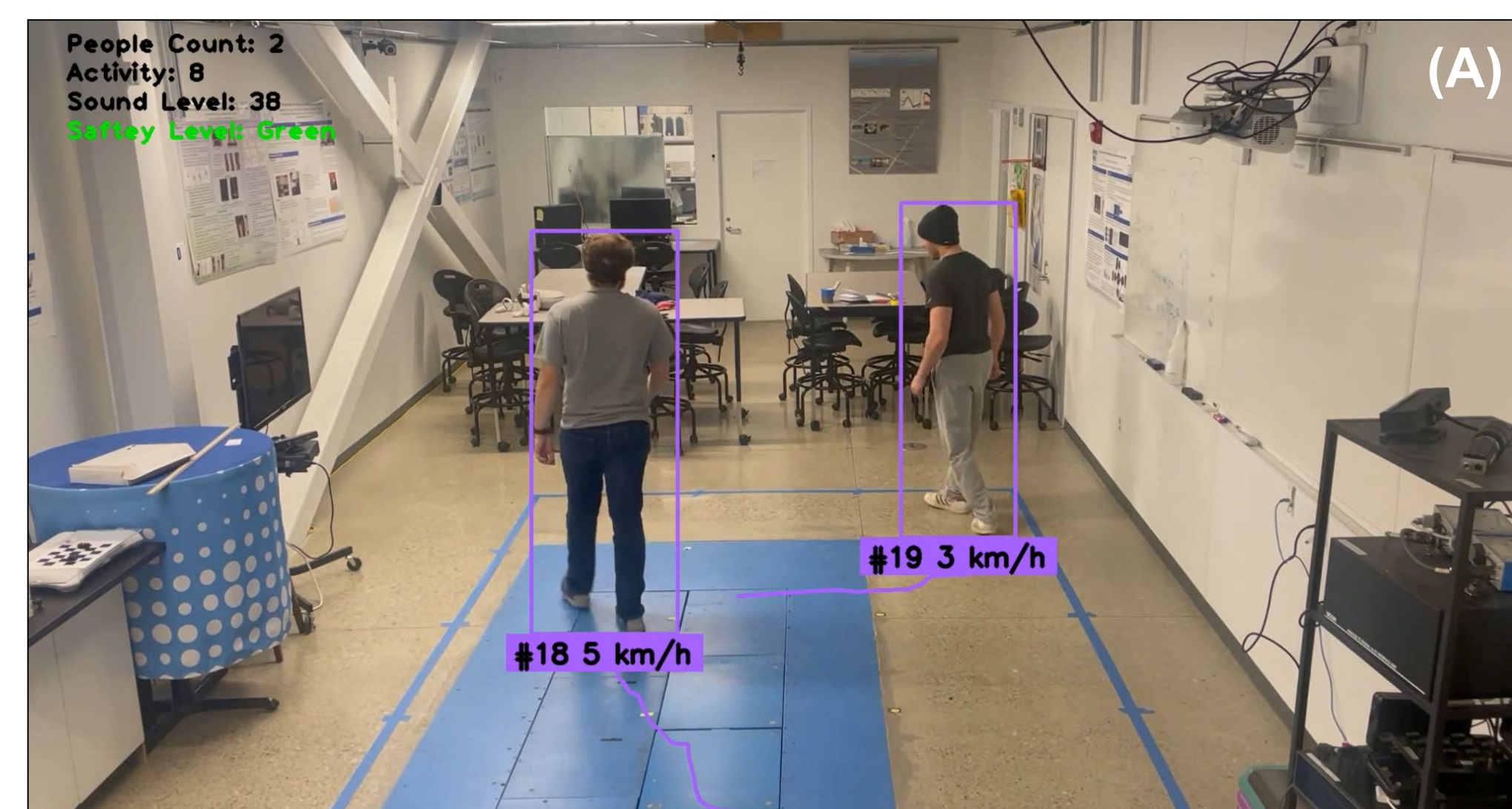


Figure 2. Algorithm functional diagram.

Algorithm Testing and Results

- A test video was shot testing with varying numbers of people, speed, body positions, and sound
- A grid was created in order to account for perspective distortion as well as test limits of polygon zone
- Video was used to validate computer vision model as well as audio calculations
- All parameters were able to be collected using only a microphone and camera

Figure 3. (A) Annotated video example; (B) Number of people over time; (C) Individual person speed over time; (D) RMS of audio over time.



Safety Factor Testing and Results

- Test videos demonstrating various scenarios were recorded
 - People seated in waiting room with limited movement around room
 - People seated and walking around the waiting room
 - Group of people entering the waiting room
 - Verbal fight/encounter with gradual escalation
 - Physical fight/encounter between two individuals and multiple people
- A grid was mapped out and taped on the floor to account for the perspective distortion of the camera and indicate the zone where data could be collected during the simulated scenarios

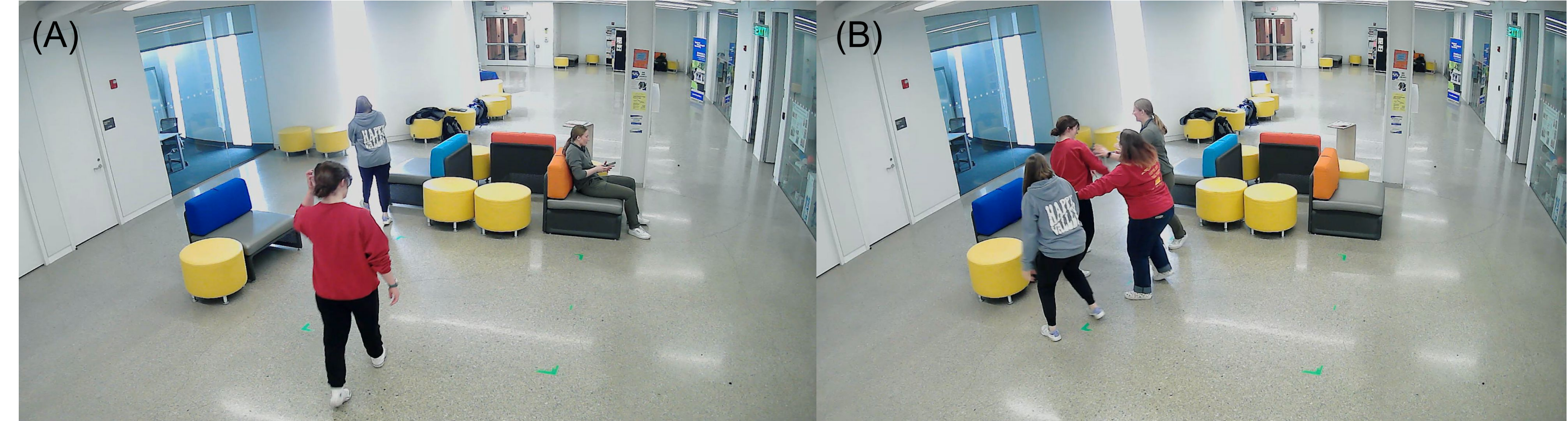


Figure 4. (A) People sitting and walking example; (B) Physical altercation example.

- The videos were recorded with a camera and its built in microphone
- Recorded videos were run through the algorithm to obtain the raw data for people count, speed of individuals, and sound
- The raw data was analyzed and used to build the equations used to calculate the safety factor score for each parameter and for the weighting of each parameter in the overall Safety Factor score calculation

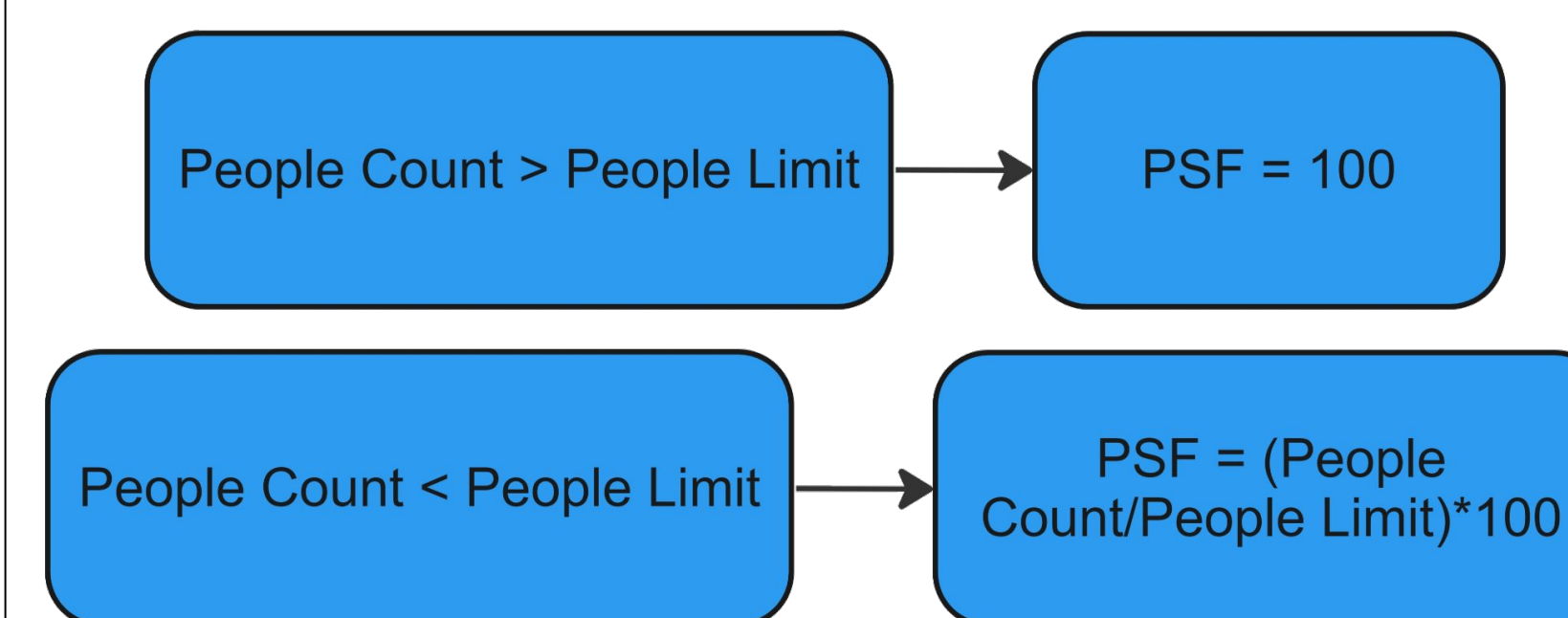


Figure 5. People safety factor calculation.

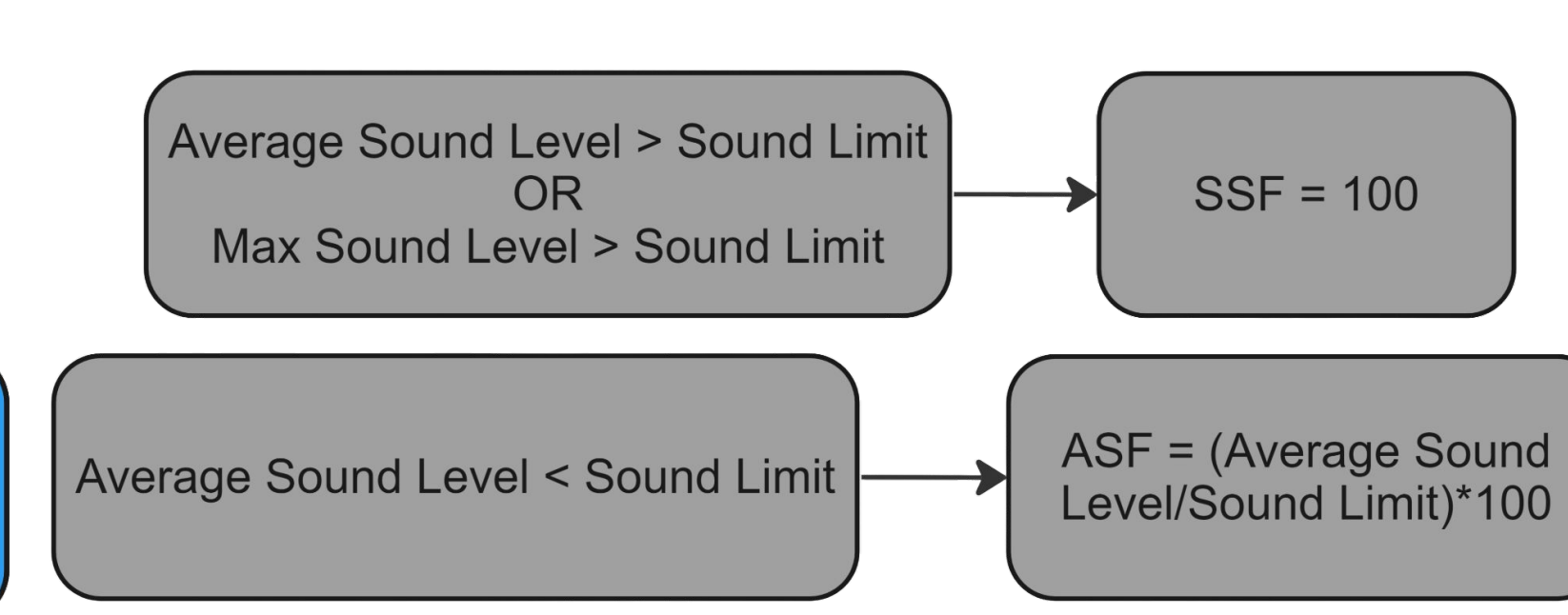


Figure 6. Sound safety factor calculation.

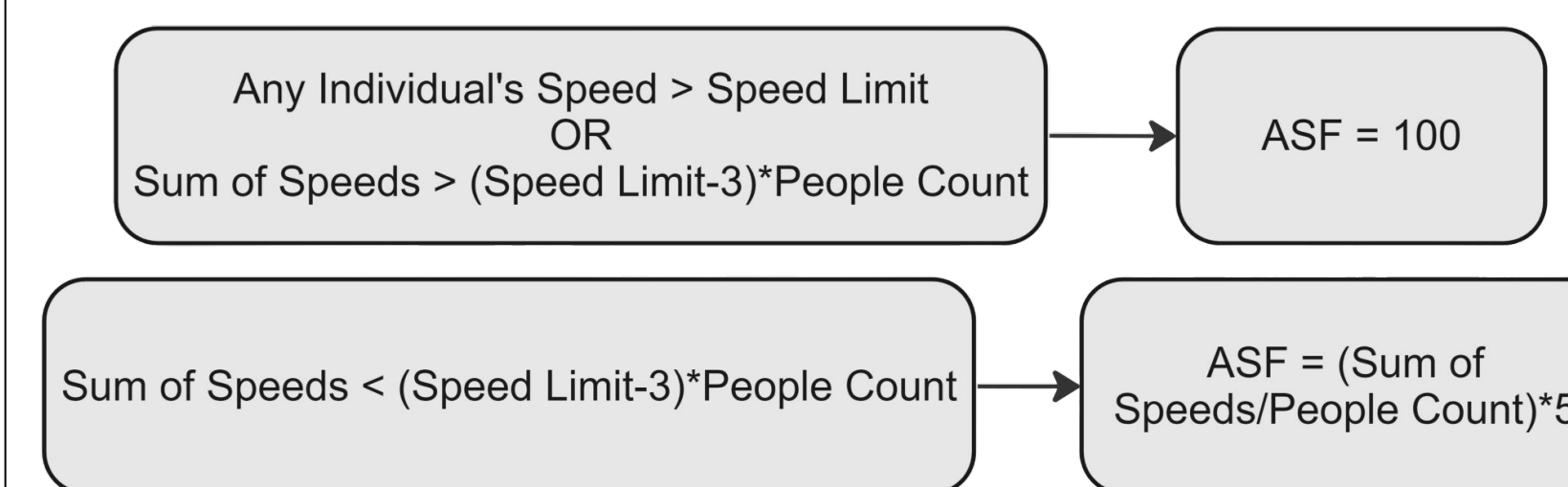


Figure 7. Activity safety factor calculation.



Figure 8. Overall safety factor calculation.

Conclusion and Future Work

Healthcare workers are facing violence in the workplace at a higher rate than ever before. Our algorithm can integrate into existing hospital security systems. The algorithm uses computer vision with object detection and tracking to count the number of people in a room, to track the speed at which people are moving, and to track the sound level in a room. Testing scenarios provided data that was used to create the Safety Factor calculation. Hospital security can use the outputted Safety Factor scores to better allocate their resources, allowing them to stop violent incidents from escalating further and prevent workplace violence altogether.

Future Work

- Get data from multiple rooms and display the Safety Factor scores on a floor plan
- Whole hospital security system
- Adapt the algorithm to be used for prison systems and schools
- Use machine learning for Safety Factor calculation
- Create an app that can send alerts to all affected hospital staff

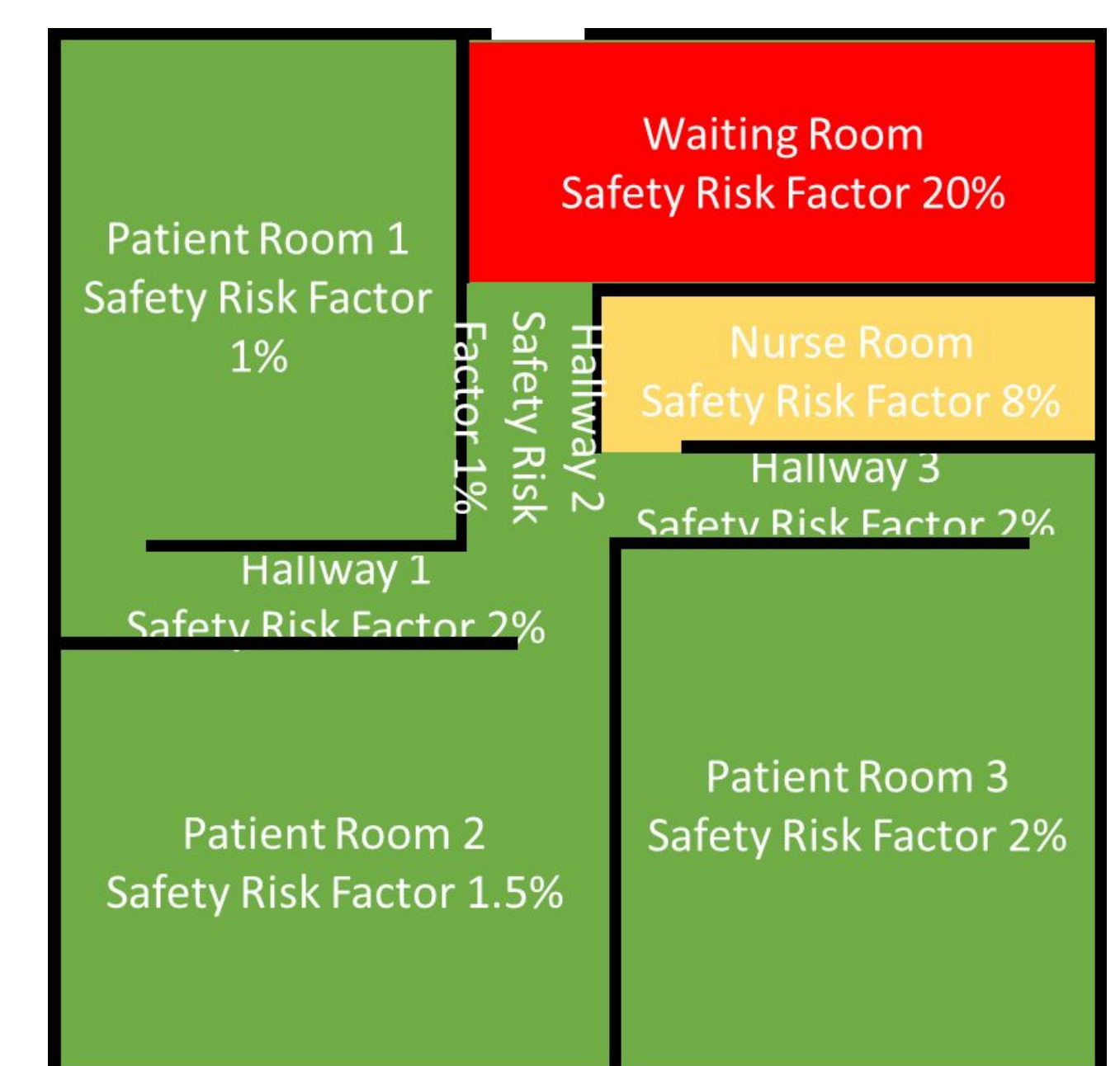


Figure 8. Floor plan with Safety Factor score mockup.

Acknowledgements

We would like to thank Lawrence Technological University and the Biomedical Engineering Department for their support of this project. We would also like to thank Dr. Eric G. Meyer and Andrew Ulaszek.

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[1] Boyle, P. (2022, August 18). *Threats against health care workers are rising, here's how hospitals are protecting their staffs*. AAMC. <https://www.aamc.org/news/threats-against-health-care-workers-are-rising-heres-how-hospitals-are-protecting-their-staffs>.
[2] *22 healthcare security, Safety and emergency management ...* International Association for Healthcare Security and Safety. (2022, March 2). <https://www.iahss.org/news/597509/22-Healthcare-Security-Safety-and-Emergency-Management-Trends-for-2022-and-Beyond.htm>.